

PERSONAL & CONTACT INFORMATION	CYENS Centre of Excellence Dimarchias Square 23 Nicosia 1016 Cyprus	Email: m.averkiou@cyens.org.cy Phone: +357-99887483 Web: https://melinos.github.io Google Scholar Profile: https://goo.gl/iXtTsn
EDUCATION	University College London, London, UK Department of Computer Science PhD in Computer Science Thesis: Data-driven Modelling of Shape Structure Advisor: Prof. Niloy Mitra Stanford University, Palo Alto, USA Geometric Computing Group Visiting PhD student Advisors: Prof. Leonidas Guibas & Dr Vladimir Kim University of Cambridge, Cambridge, UK Computer Laboratory MPhil in Advanced Computer Science (GPA: 75/100) Thesis: 3D Interfaces for 3D Modelling Advisor: Prof. Neil Dodgson University of Cyprus, Nicosia, Cyprus Department of Computer Science BSc in Computer Science (GPA: 9.3/10), <i>summa cum laude - top of the class</i> Thesis: A multi-touch interface for 3D navigation inside the virtual world of a museum exhibit Advisor: Prof. Yiorgos Chrysanthou	10/2011 – 09/2015 08/2014 – 09/2014 10/2009 – 10/2010 09/2005 – 06/2009
PAST & PRESENT EMPLOYMENT	CYENS Centre of Excellence, Nicosia, Cyprus Research Department Associate (Research) Professor Group leader of the Visual Computing Group (VCG), working on AI, with emphasis on machine learning, generative modeling, and computer vision. Results published in top venues including CVPR, ICCV and SIGGRAPH. In charge of the organization’s AI infrastructure, including Prometheus, the first peta-scale supercomputer in Cyprus dedicated to AI. MindXs Ltd, Nicosia, Cyprus Co-founder and Director Director of the R&D effort on artificial intelligence for automating electroencephalogram (EEG) signal processing for diagnostic purposes. In charge of CerebroEEG, the company’s automated EEG analysis web platform. Responsible for raising funding, secured €100k pre-seed funding. University of Cyprus, Nicosia, Cyprus Department of Computer Science Adjunct Research Scientist	07/2019 – present 03/2021 – 07/2026 07/2019 – present

Principal investigator for the ANNFASS project (2018–2021), funded by the Cyprus Research & Innovation Foundation with €250k. [ANNFASS](#) is the first neural network framework and web platform for understanding the structure and style of historical buildings.

Course instructor for Deep Learning (DSC515), Computer Vision (CS447), Advanced Computer Vision (MAI644 – part of Master in AI).

University of Cyprus, Nicosia, Cyprus

10/2015 – 06/2019

Department of Computer Science

Research Scientist

Post-doctoral research fellow, working on deep learning for 3D vision, funded by a UCY Post-Doctoral Fellowship with €44k. Results published in CVPR (**495 citations to date**), the [highest impact scientific conference](#) according to Google Scholar.

Course instructor for Visual Computing (CS607), Computer Games Engineering (CS653), Programming Problem Solving Methods (CS032).

Shenzhen Institutes of Advanced Technology, Shenzhen, China

10/2015 & 01/2016

Visual Computing Research Center

Visiting Research Scientist

Post-doctoral visitor, collaborating with local researchers on representation learning for shape analysis. Results published in ACM TOG, the top computer graphics journal (**impact factor 9.5**).

AWARDS & PRIZES

- University of Cyprus Post-Doctoral Fellowship – €44.000 2016 – 2018
- British Computer Society Distinguished Dissertation Competition, nominated 2016
- Eurographics Award for Best PhD Thesis, nominated – short-listed 2016
- University College London Studentship Award (Funded by EPSRC) – £86.000 2011 – 2015
- Eurographics Best Paper Award, 2nd prize 2014
- Rabin Ezra Scholarship Trust Bursary – £5.000 2014
- Cyprus State Scholarship Foundation PhD Scholarship – €15.000 2011 – 2014
- A.G. Leventis Foundation – PhD Scholarship £5.000 2011 – 2012
- Cyprus RIF *Students in Research* Competition, 1st prize – €3.500 2010
- University of Cambridge Studentship Award (Funded by EPSRC) – £15.500 2009 – 2010
- Cyprus State Scholarship Foundation MSc Scholarship – €5.000 2009 – 2010
- A.G. Leventis Foundation MSc Scholarship – £5.000 2009 – 2010
- Darwin College Cambridge Bursary – £1.000 2009
- Youth Board of Cyprus Postgraduate Award – €1.700 2009
- Cyprus RIF *Students in Research* Competition, 2nd prize – €3.400 2009
- Highest GPA in Univ. of Cyprus Computer Science Dept. (three awards) – €3.400 2009

GRANTS

- Horizon EuroHPC *HORIZON-EUROHPC-JU-2025-AIFA-01* – €1.000.000 2026 – present
Principal Investigator for Pharos-CY Project on behalf of CYENS.
Total project budget is €6.000.000.

SUPERVISION	Cyprus Research & Innovation Foundation <i>Infrastructures</i> Grant – €75.000 Principal Investigator for CRAFT-C Project on behalf of CYENS. Total project budget is €600.000.	2024 – present
	CYENS Centre of Excellence Internal Research Grant – €50.000 Principal Investigator for DeepRecNet Project.	2022 – 2023
	Cyprus Research & Innovation Foundation <i>PRE-SEED</i> Grant – €100.000 Principal Investigator for MindXs project.	2021 – 2022
	CYENS Centre of Excellence Seed Grant – €190.000 Principal Investigator for VCG Group.	2019 – 2022
	Cyprus Research & Innovation Foundation <i>EXCELLENCE</i> Grant – €250.000 Project Coordinator for ANNFASS Project.	2018 – 2021
	NVIDIA GPU Grant – €2.500 & €1.500	2017 & 2018
	Post-Doctoral Researchers	
	Marios Loizou (UCY)	2024 – present
	PhD Students	
	Bodhiswatta Chatterjee (Concordia University - co-supervised)	2021 – present
	Yeshwanth Kumar (CUT)	2021 – 2026
	Yiangos Georgiou (UCY)	2020 – 2025
	Marios Loizou (UCY)	2018 – 2024
	MSc Students	
	Mariam SaadAllah (UCY - Erasmus)	2025
	Luna Odenwald (UCY - co-supervised)	2025
	Menghang Hao (UCL)	2022
	Yicheng Zhan (UCL)	2022
	Jiamin Wang (UCL)	2021
	Maria Maslioukova (UCY - distinction)	2021
	Kyriakos Zantis (UCY - distinction)	2019
	Sergios Stamatis (UCY - distinction)	2018
	BSc Students	
	Panagiotis Demetriou (UCY)	2025 – present
	Andreas Mylidonis (UCY)	2021
	Stephanos Kyriakides (UCY)	2018
	Christos Othonos (UCY - distinction)	2017
	Interns	
	Tejas Anvekar	Year-round 2023-2024
	Nima Alizadeh	Summer 2022
	Andreas Kouzelis	Summer 2021
	Manos Papageorgiou	Summer 2020

- Pardis Ghaz Year-round 2020
- Mona Hodaei Year-round 2020
- Elie El Hachem Summer 2019

TEACHING

University of Cyprus, Nicosia, Cyprus

Department of Computer Science

Instructor for the following courses:

- MAI644 – Computer Vision (Master in AI) Fall 2023–2024
- DSC515 – Deep Learning Fall 2022
- CS447 – Computer Vision Spring 2021–2022–2025
- CS653 – Computer Games Software Engineering Spring 2016–2019
- Computer Games Summer School Summer 2016
- CS607 – Visual Computing Fall 2015
- CS032 – Programming Problem Solving Methods Fall 2015

University College London, London, UK

Department of Computer Science

Teaching Assistant for Image Processing (GV12) course Fall 2012–2013

PROFESSIONAL

ACTIVITIES

Program Committees

- Eurographics - Posters Chair 2024
- CVPR Workshop on Structural and Compositional Learning on 3D Data 2023
- Shape Modeling International 2018–202
- International Symposium on Visual Computing 2018–2022
- Computer Graphics International 2018–2019
- WSCG Conference on Computer Graphics, Visualization and Computer Vision 2017
- IEEE Melecon 2016
- SIGGRAPH Asia Workshop on Creative Shape Modeling and Design 2014

Reviewer in International Journals

ACM Transactions on Graphics, Computer Graphics Forum, IEEE Transactions on Visualization and Computer Graphics, Computers and Graphics, Graphical Models, Computer Graphics and Applications, Journal of Artificial Intelligence Research, Knowledge-Based Systems

Reviewer in International Conferences

CVPR, ICCV, ECCV, WACV, SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, Computer Graphics International, Shape Modeling International, International Symposium on Visual Computing

ADMINISTRATIVE DUTIES	CYENS Centre of Excellence, Nicosia, Cyprus AI Cluster Committee Chair 2021 – present – Led strategic planning for AI research across the organization – Developed policies for shared computing resource allocation – Coordinated cross-group collaborations involving 20+ researchers Prometheus HPC Initiative – Project Lead 2019 – present – Co-wrote technical specifications for CYENS’ peta-scale AI supercomputer – Led procurement process for €500k computing infrastructure – Co-designed resource allocation and scheduling policies – Represent CYENS at EU PRACE organization for HPC since 2024 Doctoral Training Program Committee Chair 2020 – 2022 – Developed the program’s policies including candidate selection, evaluation and progress monitoring – Chaired several selection cohorts evaluating dozens of applicants to the fellowship program – Organized evaluation committees for monitoring fellows’ progress Scientific Council Member 2019 – present – Contributed to organization-wide research strategy – Contributed to developing policies for the Research Department – Participated in several hiring committees including IT Manager, 3D Developers, and junior Research Group Leaders
JOURNAL PUBLICATIONS	[1] Bartol, K., Hentschel, F., Sadretdinova, N., Russig, B., Averkiou, M., Kyosev, Y., Gumhold, S., “EASE: Parametric garment design with explicit and local ease control,” <i>Computers & Graphics</i>, vol. 138, p. 104644, Jun. 2026, Article. DOI: 10.1016/j.cag.2026.104644. [2] Georgiou, Y., Loizou, M., Averkiou, M., Kalogerakis, E., “Im2SurfTex: Surface Texture Generation via Neural Backprojection of Multi-View Images,” <i>Computer Graphics Forum</i>, vol. 44, no. 5, 2025, also presented in SGP 2025. Project Website. [3] Othonos, C., Besuievsky, G., Averkiou, M., Pelecanos, L., Patow, G., Chrysanthou, Y., “Procedural modelling of aged buildings: A case study of mudbrick houses in cyprus,” <i>ACM Journal on Computing and Cultural Heritage</i>, vol. 18, no. 4, 2025. [4] Loizou, M., Garg, S., Petrov, D., Averkiou, M., Kalogerakis, E., “Cross-Shape Attention for Part Segmentation of 3D Point Clouds,” <i>Computer Graphics Forum</i>, vol. 42, no. 5, 2023, also presented in SGP 2023. Project Website. [5] Artopoulos, G., Maslioukova, M. I., Zavou, C., Loizou, M., Deligiorgi, M., Averkiou, M., “An Artificial Neural Network Framework for annotating and classifying Architectural Structure and Style of Built Heritage in 3D,” <i>Journal of Cultural Heritage</i>, vol. 63, pp. 135–147, 2022, Project Website. [6] Deligiorgi, M., Maslioukova, M. I., Averkiou, M., Andreou, A., Selvaraju, P., Kalogerakis, E., Patow, G., Chrysanthou, Y., Artopoulos, G., “A 3D digitisation workflow for architecture-specific annotation of built heritage,” <i>Journal of Archaeological Science: Reports</i>, vol. 37, 102787:1–102787:11, 2021, Project Website.

- [7] Loizou, M., **Averkiou, M.**, Kalogerakis, E., “Learning Part Boundaries from 3D Point Clouds,” *Computer Graphics Forum*, vol. 39, no. 5, pp. 183–195, 2020, also presented in SGP 2020 - acceptance rate: 36%. [Project Website](#).
- [8] Hu, R., Li, W., Kaick, O., Huang, H., **Averkiou, M.**, Cohen-Or, D., Zhang, H., “Co-Locating Style-Defining Elements on 3D Shapes,” *ACM Transactions on Graphics*, vol. 36, no. 3, 33:1–33:15, 2017, also presented in SIGGRAPH 2017 - acceptance rate: 28%. [Project Website](#).
- [9] **Averkiou, M.**, Kim, V. G., Mitra, N. J., “Autocorrelation Descriptor for Efficient Co-alignment of 3D Shape Collections,” *Computer Graphics Forum*, vol. 35, no. 1, pp. 261–271, 2016, also presented in Eurographics 2016. [Project Website](#).
- [10] **Averkiou, M.**, Kim, V. G., Zheng, Y., Mitra, N. J., “ShapeSynth: Parameterizing Model Collections for Coupled Shape Exploration and Synthesis,” *Computer Graphics Forum*, vol. 33, no. 2, pp. 125–134, 2014, also presented in Eurographics 2014 - acceptance rate: 25%. [Project Website](#).
- [11] *Fish, N., ***Averkiou, M.**, Kaick, O., Sorkine-Hornung, O., Cohen-Or, D., Mitra, N. J., “Meta-representation of Shape Families,” *ACM Transactions on Graphics*, vol. 33, no. 4, 34:1–34:11, 2014, also presented in SIGGRAPH 2014 - acceptance rate: 25%. *joint first authors [Project Website](#).
- [12] Zheng, Y., Cohen-Or, D., **Averkiou, M.**, Mitra, N. J., “Recurring Part Arrangements in Shape Collections,” *Computer Graphics Forum*, vol. 33, no. 2, pp. 115–124, 2014, also presented in Eurographics 2014 - acceptance rate: 25% - **Best Paper Award, 2nd prize**. [Project Website](#).
- [13] Adimoolam, Y. K., Poullis, C., **Averkiou, M.**, “Data Leakage Detection and De-duplication in Large Scale Geospatial Image Datasets,” in *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, [Project Website](#).
- [14] Adimoolam, Y. K., Poullis, C., **Averkiou, M.**, “Pix2Poly: A Sequence Prediction Method for End-to-end Polygonal Building Footprint Extraction from Remote Sensing Imagery,” in *Proceedings of Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 8473–8472, [Project Website](#).
- [15] Petrov, D., Goyal, P., Shivashok, D., Tao, Y., **Averkiou, M.**, Kalogerakis, E., “ShapeWords: Guiding Text-to-Image Synthesis with 3D Shape-Aware Prompts,” in *Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, [Project Website](#).
- [16] Georgiou, Y., Loizou, M., Kelly, T., **Averkiou, M.**, “FacadeNet: Conditional Facade Synthesis via Selective Editing,” in *Proceedings of Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 5384–5393, [Project Website](#).
- [17] Petrov, D., Goyal, P., Thamizharasan, V., Kim, V. G., Gadelha, M., **Averkiou, M.**, Chaudhuri, S., Kalogerakis, E., “GEM3D: Generative Medial Abstractions for 3D Shape Synthesis,” in *Proceedings of SIGGRAPH*, 2024, 92:1–92:11, [Project Website](#).
- [18] Georgiou, Y., **Averkiou, M.**, Kelly, T., Kalogerakis, E., “Projective Urban Texturing,” in *Proceedings of International Conference on 3D Vision (3DV)*, 2021, pp. 1034–1043, [Project Website](#).
- [19] Selvaraju, P., Nabail, M., Loizou, M., Maslioukova, M. I., **Averkiou, M.**, Andreou, A., Chaudhuri, S., Kalogerakis, E., “BuildingNet: Learning to Label 3D Buildings,” in *Proceedings of IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10397–10407, **Oral Paper - acceptance rate: 3%**. [Project Website](#).

CONFERENCE
PUBLICATIONS

- [20] Lin, H., **Averkiou, M.**, Kalogerakis, E., Kovacs, B., Ranade, S., Kim, V. G., Chaudhuri, S., Bala, K., “Learning Material-Aware Local Descriptors for 3D Shapes,” in *Proceedings of International Conference on 3D Vision (3DV)*, 2018, pp. 150–159.
- [21] Kalogerakis, E., **Averkiou, M.**, Maji, S., Chaudhuri, S., “3D Shape Segmentation with Projective Convolutional Networks,” in *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3779–3788, **Oral Paper - acceptance rate: 2.65%.** [Project Website](#).
- [22] Zheng, S., Prisacariu, V. A., **Averkiou, M.**, Cheng, M.-M., Mitra, N. J., Shotton, J., Torr, P. H., Rother, C., “Object Proposals Estimation in Depth Image Using Compact 3D Shape Manifolds,” in *Lecture Notes in Computer Science*, ser. Proceedings of German Conference on Pattern Recognition (GCPR), vol. 9358, 2015, pp. 196–208, [Project Website](#).
- [23] **Averkiou, M.**, Mitra, N. J., “Automatic Alignment of Shape Collections,” in *Proceedings of Eurographics 2012 - Posters Track*, 2012.
- [24] **Averkiou, M.**, Dodgson, N., “Comparison of relative (mouse-like) and absolute (tablet-like) interaction with a large stereoscopic work-space,” in *Proceedings of the Stereoscopes Displays and Applications XXII Conference*, 2011.
- [25] **Averkiou, M.**, Chrysanthou, Y., “Evaluating a multi-touch interface for 3D navigation inside the virtual world of a museum exhibit,” in *Proceedings of the 10th VAST International Symposium on Virtual Reality, Archaeology and Cultural Heritage*, 2009.
- [26] Kunkel, T., **Averkiou, M.**, Chrysanthou, Y., “A web-based virtual museum application,” in *Proceedings of the 14th International Conference on Virtual Systems and Multimedia*, 2008.

THESES & TECHNICAL REPORTS

- [27] **Averkiou, M.**, *Data-driven Modelling of Shape Structure*. University College London, 2015.
- [28] **Averkiou, M.**, “3D Interfaces for 3D Modelling,” University of Cambridge, Tech. Rep., 2010.
- [29] **Averkiou, M.**, “Digital Watermarking,” University of Cambridge, Tech. Rep., 2010.
- [30] **Averkiou, M.**, “A multi-touch interface for 3D navigation inside the virtual world of a museum exhibit,” University of Cyprus, Tech. Rep., 2009.

RESEARCH IMPACT

h-index: 15
i10-index: 16
Citations: 1175
as of 24/08/2026

TALKS & PRESENTATIONS

- Eurographics 04/2014, 05/2016
- Shenzhen Institutes of Advanced Technology (invited) 10/2015
- Stanford Geometric Computing Group (invited) 09/2014
- SIGGRAPH 08/2014

INDUSTRY PARTNERSHIPS

MindXs Ltd 2021 – 2026
Co-founded health-tech startup applying AI to medical signal processing; developed data analytics pipeline for EEG recordings; secured €100k pre-seed funding

Adobe Research 2015 – present
Long-term research collaboration on 3D shape analysis and synthesis; co-authored publications

with Adobe researchers including Dr. Vladimir Kim and Dr. Siddhartha Chaudhuri; explored applications of research to Adobe's 3D modeling tools

VAIVR / Somata Labs

2024 – present

Developed a neural network for extracting human body measurements from a single photo and converting them to a 3D avatar; designed the data pipeline for the company's first prototype

Altair Engineering

2022

Consulting project on AI-driven 3D semantic segmentation; developed prototype for CAD tools; knowledge transfer to HyperWorks product engineering team

Shape Labs

2019

Technical consulting on neural-based 3D correspondence matching for human shapes; advisory role on data architecture for processing large-scale 3D human scans acquired via photogrammetry

NVIDIA

2017 & 2018

Secured GPU donations through competitive academic grants on two separate occasions for training neural networks

RESEARCH

DATA & CODE

Datasets Created and Maintained

- **BuildingNet**

2021

First large-scale dataset of 2,000+ 3D building models with fine-grained semantic part annotations; published as an oral paper at ICCV 2021; part of benchmark challenge in CVPR 2023 Workshop on Structural and Compositional Learning on 3D Data; used by dozens of research groups worldwide

- **ANNFASS Cypriot Heritage Collection**

2020

Dataset of historical buildings with architectural style annotations and structural analysis; developed as part of €250k Excellence grant from Cyprus Research & Innovation Foundation

Data Platforms and Tools

- **ShapeWords Demo**

2025

Online demo of the ShapeWords generative image model conditioned on 3D shapes, hosted in Hugging Face

- **BuildingNet Editor**

2021

Web platform for architectural data mining and annotation

- **ANNFASS Platform**

2021

Online platform for architectural heritage analysis supported by neural networks

- 10+ code repositories on Github with open source and data in the past 5 years

TECHNICAL

SKILLS

High-Performance Computing: Conceptualized, specified, procured, and manage Prometheus AI supercomputer: 2.14 PFLOPs FP32, 17.13 PFLOPs FP16, InfiniBand 200G HDR network, 300TB LustreFS parallel filesystem; experience with Slurm workload manager; familiar with CUDA optimization for deep learning

Big Data Technologies: PyTorch torchrun for large-scale model training; developed custom data loaders and pre-processing pipelines for terabyte-scale datasets including Objaverse (100+ TB),

ShapeNet (5+ TB), and TUH-EEG (8+ TB); pandas and NumPy for data manipulation; experience with parallel and distributed data processing

AI & Deep Learning: Advanced PyTorch implementation of CNNs (ResNet, EfficientNet), GNNs (BuildingGNN), and transformers (Cross-shape attention); experienced in optimizing diffusion models (GEM3D) and GANs (FacadeNet)

Data Analysis: Experience with 3D data processing algorithms using Open3D; experience with point cloud libraries (PCL); scikit-learn for machine learning applications; data visualization with Matplotlib and three.js; demos with gradio and flask